The Disagreement Problem:

How Model Generalization Impacts XAI Consensus

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Motivation

- XAI methods frequently **disagree** on feature importance for the same model and input.
- Prior work (Krishna et al., 2024) treats disagreement as a mathematical artifact of the methods themselves.
- **Our hypothesis:** disagreement is also a symptom of *model quality* — poorly generalising models produce unreliable gradient landscapes that amplify or mask inter-method divergence.

Research Question

Does model generalisation quality causally drive XAI consensus or disagreement, independent of mathematical alignment?

Dataset & Model Spectrum

Dataset: UCI Adult Income — binary classification, 105 features after one-hot encoding.

Four models across a degradation spectrum:

Label	Model Type	Corruption	Role
A	MLP (Control)	None	Baseline
C	MLP + Feat. Noise	Gaussian input	Mild degradation
D	MLP + Label Poison	20% label flip	Severe degradation
B	MLP (Overfit)	Memorised train set	Worst generalisation

XAI Methods & Alignment

Four methods applied uniformly:

- **LIME** — local linear approximation (perturbation-based)
- **KernelSHAP** — Shapley-value kernel estimator (perturbation-based)
- **Integrated Gradients (IG)** — path-integral attribution (gradient-based)
- **SmoothGrad** — noise-averaged gradient (gradient-based)

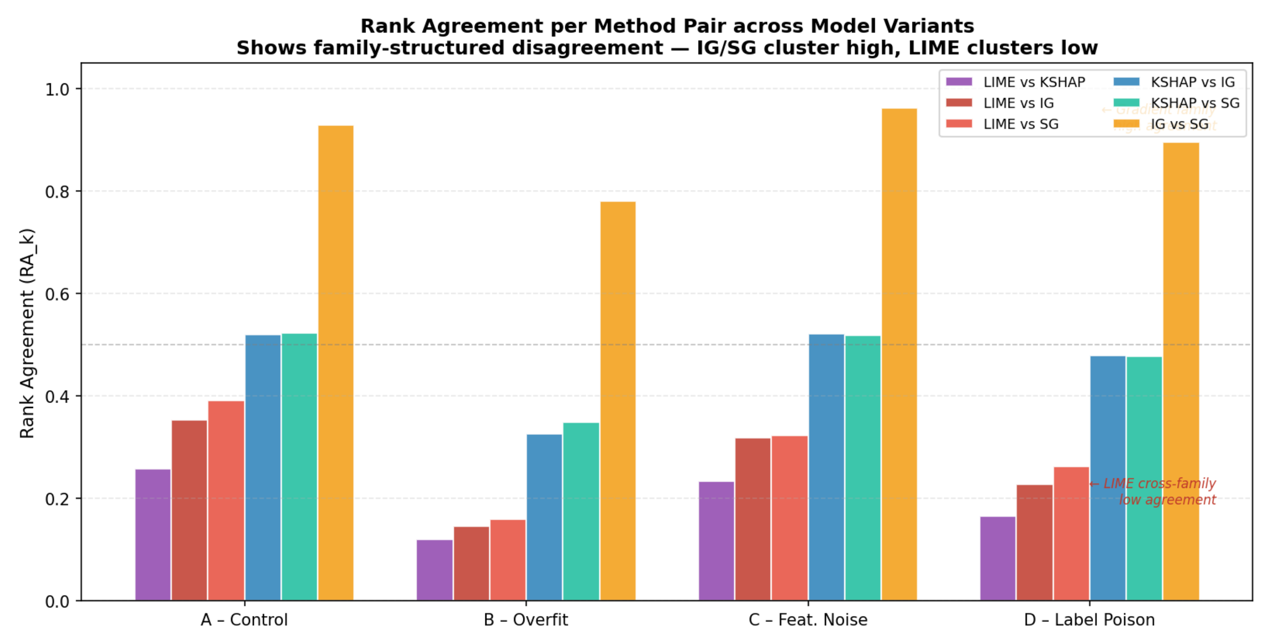
Mathematical alignment: all attributions projected onto the *same* 105 feature indices, eliminating index-mismatch as a confound (Krishna et al., 2024).

Disagreement Metrics

Three pairwise scores across all $\binom{4}{2} = 6$ method pairs:

- **Rank Agreement (RA):** Spearman correlation of importance ranks.
- **Sign Agreement (SA):** Fraction of features with matching attribution sign.
- **Signed Rank Agreement (SRA):** Combines rank *and* sign — our primary metric.

$$\mathrm{SRA}(e_1, e_2) = \frac{1}{|F|} \sum_{f \in F} \mathbf{1}[\mathrm{rank}_{e_1}(f) = \mathrm{rank}_{e_2}(f)] \cdot \mathbf{1}[\mathrm{sign}_{e_1}(f) = \mathrm{sign}_{e_2}(f)]$$



Finding 1 — Overfitting Destroys Consensus

- As generalisation degrades (A→C→D→B), mean pairwise SRA **monotonically decreases**.
- SRA: **0.734** (Model A) → **0.649** (Model B) — an 11.6% relative decline.
- RA and SA follow the same trend, confirming result is not metric-specific.

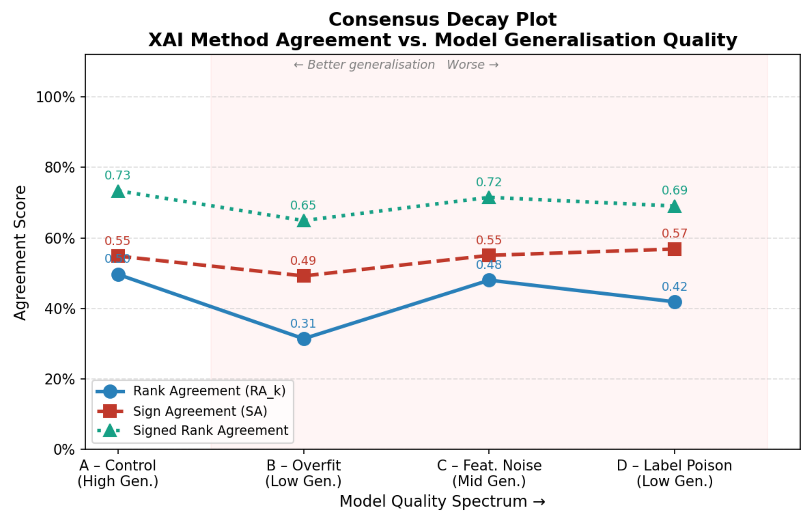


Fig. 1a: Agreement vs. model accuracy (RA / SA / SRA lines).

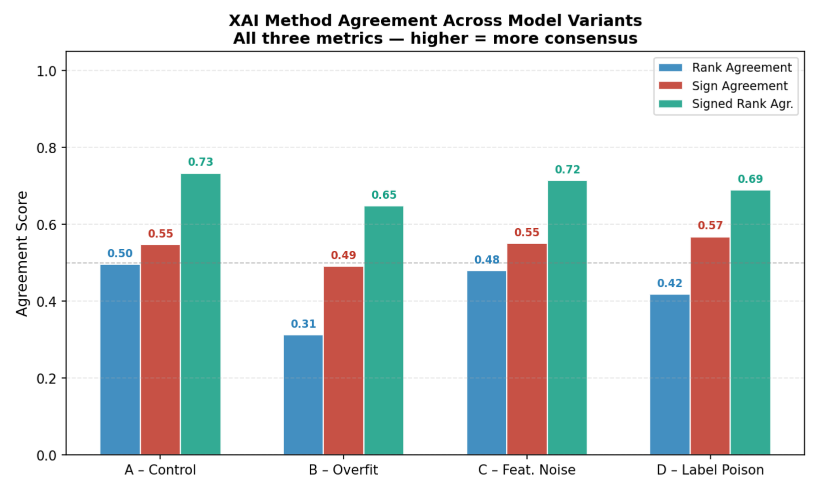


Fig. 1b: RA / SA / SRA breakdown across the four models.

Metric	A (Control)	C (Noise)	D (Poison)	B (Overfit)
SRA	0.734	0.72	0.69	0.649

Finding 2 — The Uncertainty Paradox

- **Counterintuitive:** predictive entropy *positively* correlates with SRA.
- Degraded models produce **flat gradient landscapes** — near-zero attributions everywhere, so methods trivially agree on a "nothing matters" explanation.
- This is *surface-level consensus*: agreement is high, explanations are **uninformative**.
- SRA alone cannot certify quality; **entropy must be reported alongside**.

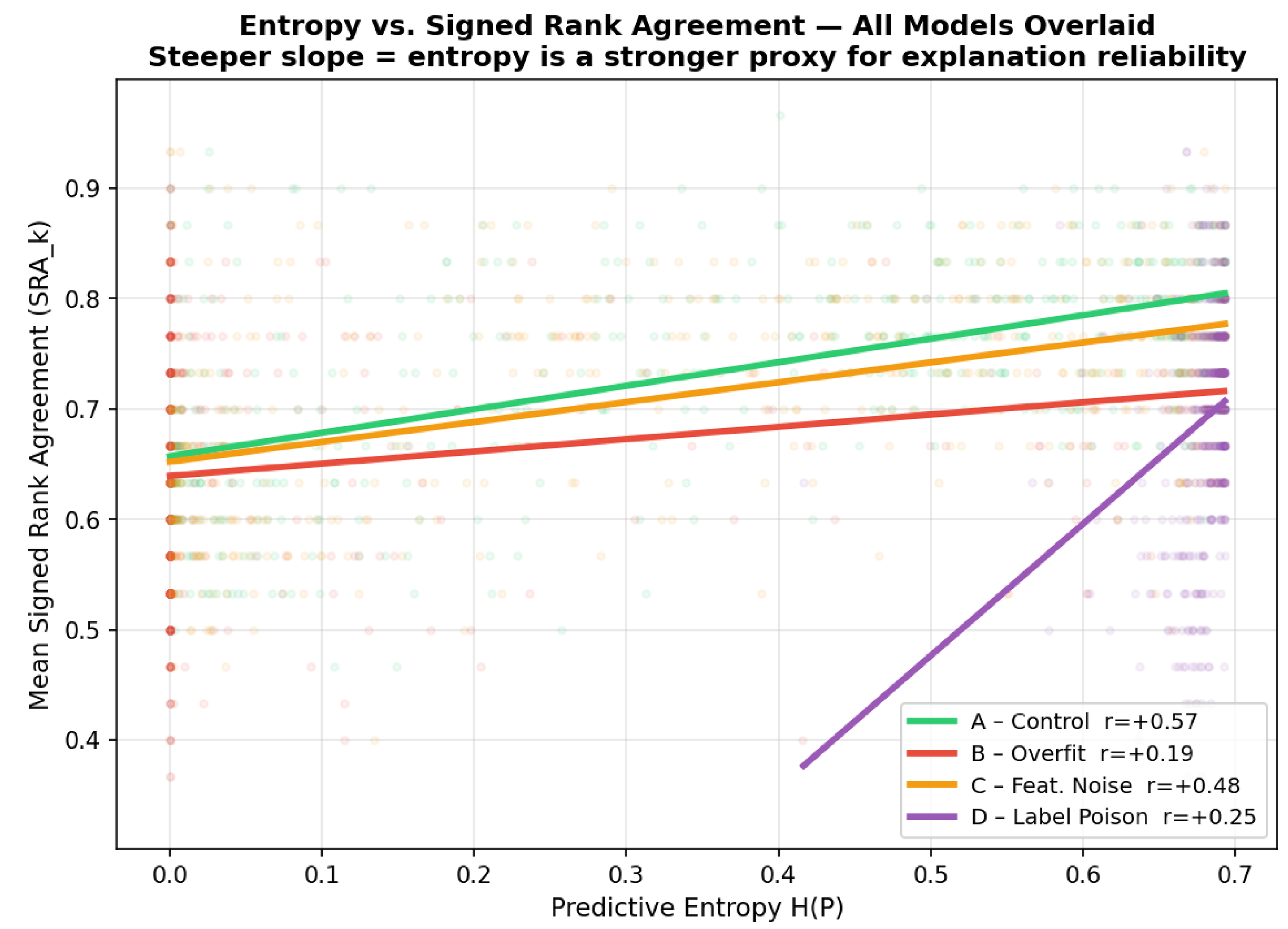


Fig. 2: Predictive entropy vs. mean pairwise SRA (coloured by model). Positive trend confirms the paradox.

Finding 3 — Family-Structured Disagreement

- XAI disagreement is **not uniform**: methods split into two families.
- **Gradient family** (IG + SmoothGrad): high mutual agreement across all models.
- **Perturbation family** (LIME): consistently isolated; largest divergence.
- **KernelSHAP** occupies a middle ground, shifting allegiance under degradation.
- In degraded models, SHAP and SmoothGrad **contradict each other on class direction** (SA $= 0.343$).

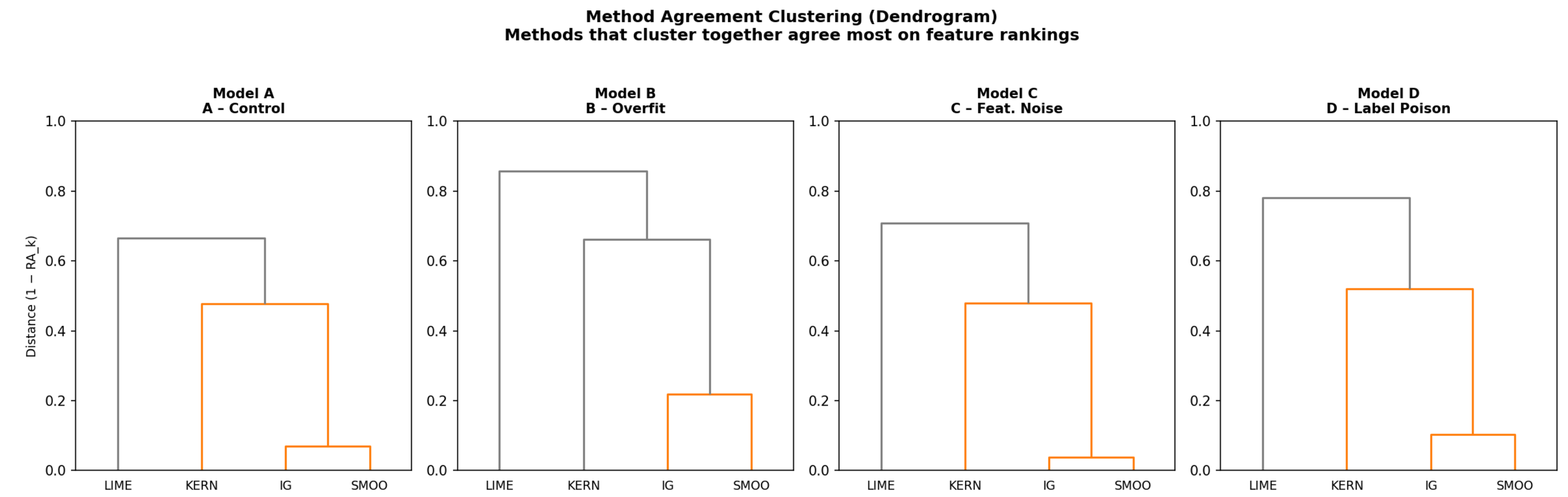


Fig. 3: Radar/dendrogram of pairwise agreement — gradient-method clustering and perturbation-method isolation.

Pair	Family	SA (Model B)
IG – SmoothGrad	Gradient–Gradient	≈ 0.89
SHAP – SmoothGrad	Perturbation–Gradient	0.343
LIME – IG	Perturbation–Gradient	low

Critical Caveat: Consensus $\neq$ Faithfulness

- IG and SmoothGrad agree due to *shared mathematical assumptions* (gradient linearity), not ground-truth correctness.
- Intra-family consensus is **evidence of shared bias**, not validation.
- Practitioners must not use agreement as a proxy for correctness.

Contributions

1. First empirical demonstration that **model generalisation quality** drives XAI consensus, independent of mathematical alignment.
2. Identification of the **Uncertainty Paradox**: high entropy $\Rightarrow$ spuriously high agreement.
3. Characterisation of **family-structured disagreement** across gradient vs. perturbation methods.
4. Practical guidance: agreement metrics must be read *jointly* with entropy and method-family membership.

Conclusions

- XAI disagreement is a symptom of *model failure*, not only method incompatibility.
- Deploying XAI on overfit/miscalibrated models risks both spurious disagreement *and* spurious consensus.
- **Model quality auditing must precede XAI interpretation.**

Future Work

- Extend to image/NLP datasets and transformers/GBDTs.
- Develop entropy-conditioned XAI selection criteria.
- Investigate ensembling to decouple consensus from faithfulness.

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References

- Krishna et al. (2024) *The Disagreement Problem in Explainable Machine Learning*. ICML 2024